WEEK 3 / LESSON 6 / 50 MINUTES

L6

Database Design

Choosing SQL when you needed NoSQL — or vice versa — is the startup bug that takes 6 months to unravel. Today: a decision framework you will use for your whole career.

Which of your apps store everything in one MongoDB collection "just to ship fast"? We talk about it.

WHAT'S INSIDE

WHY THIS LESSON MATTERS

The right schema is the one you do not regret at 100× scale.

WHY

Real

A real reason this matters in industry today.

SCALE

Today

A real reason this matters in industry today.

SPEED

Now

A real reason this matters in industry today.

IMPACT

Yours

A real reason this matters in industry today.

LEARNING OUTCOMES

By the end of this lesson, you will be able to —

1. Choose SQL or NoSQL given a data shape and access pattern.
2. Design a 3-table schema in third normal form.
3. Write JOINs, aggregates, and indexes for AI workloads.
4. Model document schemas in MongoDB — embed vs reference.
5. Connect a Node.js API to both Postgres and Mongo.

AGENDA

What we'll cover today

Short, sequenced parts. Each one builds on the last.

01

SQL vs NoSQL — the one...

SQL first. Move to NoSQL only with a specific, justified reason.

02

Normalisation — the three...

Rules of thumb, not commandments

03

Indexes — where performance...

80/20 of all query-speed issues

04

MongoDB — embed vs reference

Wrong choice now = a schema migration at 100k users.

01

PART 1 OF 4

SQL vs NoSQL — the one...

SQL first. Move to NoSQL only with a specific, justified reason.

CONCEPT 1 OF 4

SQL vs NoSQL — the one decision that matters

SQL first. Move to NoSQL only with a specific, justified reason.

DEFINITION

SQL vs NoSQL — the one decision that...

SQL first. Move to NoSQL only with a specific, justified reason.

EXAMPLE

Real-world example: applies directly to Indian industry contexts.

CONCEPT 2 OF 4

Normalisation — the three forms

1NF: atomic columns — no arrays in a cell.
2NF: no partial dependencies on a composite PK.
3NF: no transitive dependencies (A → B → C).
Pragmatic: normalise to 3NF; denormalise only for measured wins.
GUIDE
3NF then stop

Premature denormalisation is the #1 startup schema bug.

KEY POINTS

- 1NF: atomic columns — no arrays in a cell
- 2NF: no partial dependencies on a composite PK
- 3NF: no transitive dependencies (A → B → C)
- Pragmatic: normalise to 3NF; denormalise only for measured wins

CONCEPT 3 OF 4

Indexes — where performance lives

B-tree index on PK is automatic.
Index every FK you filter by.
Composite index for multi-column WHERE — order matters.
Every index slows writes; audit quarterly.
SKILL
EXPLAIN

Learn EXPLAIN before you learn any other tuning knob.

DEFINITION

Indexes — where performance lives

80/20 of all query-speed issues

EXAMPLE

Real-world example: applies directly to Indian industry contexts.

CONCEPT 4 OF 4

MongoDB — embed vs reference

Wrong choice now = a schema migration at 100k users.

KEY POINTS

- Wrong choice now = a schema migration at 100k users
- Practical, named, applied to a real Indian use-case.
- Practical, named, applied to a real Indian use-case.

02

PART 2 OF 4

Engage and reflect

Pause, talk, and connect what you just learned to your own work.

✦ THINK · PAIR · SHARE

ACTIVITY

Design a proptech in 5 minutes

PROMPT

Apply the four concepts above to one real example from your own week.

HOW TO RUN IT

1.

2. Decide each: SQL or NoSQL. List the FK indexes.

3. Defend the split. Room critiques.

4. Reveal: instructor confirms the canonical answer.

REFLECT & APPLY

Three questions before you close this lesson

Spend 60 seconds on each. Write the answers down — no scrolling, no shortcuts.

Q1 *Name one project where you chose NoSQL too early.*

Q2 *Which index would you add to your current DB before Sunday?*

Q3 *Embed or reference — which is your default, and why?*

03

PART 3 OF 4

Knowledge check

Three short questions. One minute each. Honest answers only.

KNOWLEDGE CHECK · QUESTION 1 OF 3

Default schema choice for most AI startups:

A MongoDB

B DynamoDB

C PostgreSQL

D Redis

ANSWER · QUESTION 1 OF 3

Default schema choice for most AI startups:

CORRECT ANSWER

C

PostgreSQL

WHY

SQL first. Most data shapes are known; joins are valuable; transactions matter

KNOWLEDGE CHECK · QUESTION 2 OF 3

A table has customer_name in an Order row. Fix?

- **A** Nothing
- **B** Move to Customer; FK
- **C** Delete column
- **D** Index it

ANSWER · QUESTION 2 OF 3

A table has customer_name in an Order row. Fix?

CORRECT ANSWER

B

Move to Customer; FK

WHY

3NF violation — move to Customer; join on customer_id

KNOWLEDGE CHECK · QUESTION 3 OF 3

EXPLAIN reveals a sequential scan on a 10M-row table. Fix?

A Rewrite query

B Add an index

C Shard

D Upgrade DB

ANSWER · QUESTION 3 OF 3

EXPLAIN reveals a sequential scan on a 10M-row table. Fix?

CORRECT ANSWER

B

Add an index

WHY

Most slow scans are fixed by a well-chosen index

04

PART 4 OF 4

Apply and close

One thing to remember. One thing to ship this week.

SUMMARY

Five sentences to remember

If you remember nothing else from this lesson, remember these five lines.

1. Pick the right tool — never the loudest one.
2. Practice the framework on a real example.
3. Watch for the three classic pitfalls.
4. Document your decision in one sentence.
5. Carry the discipline forward to the next lesson.

WORKED EXAMPLE

How this lesson plays out in one real Indian context

THE SCENARIO

SQL first. Move to NoSQL only with a specific, justified reason.

THE TAKEAWAY

"The right schema is the one you do not regret at 100× scale."

WHAT TO NOTICE

1. SQL vs NoSQL — the one...
2. Normalisation — the three...
3. Indexes — where performance...
4. MongoDB — embed vs reference

APPLY IT → ASSIGNMENT

Mini assignment — proptech ER diagram

Design a 4-table schema for the real-estate platform used in P1.

DELIVERABLES

- Tables: users, listings, viewings, reviews.
- Identify FKs + FK indexes.
- Draw an ER diagram (any tool — excalidraw, dbdiagram.io).
- Write the CREATE TABLE script for Postgres.

WHAT 'GOOD' LOOKS LIKE

Deliverable: ER diagram PNG + CREATE TABLE .sql file on GitHub.

ONE THING TO REMEMBER

“

"The right schema is the one you do not regret at 100× scale."

— AI Learning Hub

END OF LESSON 3

What you can now do

Each one of these is now part of your working vocabulary.

1. Pick the right tool — never the loudest one.
2. Practice the framework on a real example.
3. Watch for the three classic pitfalls.
4. Document your decision in one sentence.
5. Carry the discipline forward to the next lesson.

NEXT → Lesson 7 — Intelligent Document Processing. OCR meets NLP.